ARTICLE

Longitudinal characterization of impulsivity phenotypes boosts signal for genomic correlates and heritability

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Genomic correlates of impulsivity have been identified in several genome-wide association studies (GWAS) using cross-sectional designs, but no studies have investigated the molecular genetic correlates of impulsivity phenotypes using longitudinally constructed traits. In 3860 unrelated European participants in the Avon Longitudinal Study of Parents and Children (ALSPAC), we constructed longitudinal phenotypes for delay discounting and impulsive personality traits (as measured by the UPPS-P impulsive behavior scales) via assessment at ages 24, 26, and 28. We conducted GWASs of impulsivity using both cross-sectional and longitudinal phenotypes, estimated heritability and their phenotypic and genetic correlations, and evaluated their association with recently-developed polygenic risk scores (PRSs) for the impulsivity indicators themselves and also related psychiatric conditions. Latent growth curve modeling revealed a stable intercept over time for all impulsivity phenotypes. High genetic correlation of cross-sectional measures over time suggested a stable genetic component for delay discounting ($r_g = 0.53$–0.99) and sensation seeking ($r_g = 0.99$). Heritability estimates of the stable longitudinal phenotypes substantively improved as compared to their cross-sectional counterparts, revealing a significant SNP-heritability for delay discounting (0.22; $p = 0.03$) and sensation seeking (0.35; $p = 0.0007$). Consistent with previous reports, GWAS and gene-based analyses revealed associations between specific longitudinal impulsivity indicators and *CADM2* and *NCAM1* genes. The PRSs for the impulsivity indicators and disorders related to self-regulation were also significantly associated with longitudinal impulsivity traits. Finally, we validated the associations between longitudinal impulsivity phenotypes and their PRSs in an independent 13-wave longitudinal study ($n = 1019$) and the benefit of longitudinal phenotypes in simulation studies. In this first longitudinal genetic study of impulsivity traits, the results revealed stable genomic correlates of delay discounting and sensation seeking over time and further validated the utility of recently-developed PRSs, both in relation to the observed traits and in connecting them to psychiatric disorders. More generally, these findings support using latent intercepts as novel longitudinal phenotypes to boost signal for heritability and genomic correlates of mechanisms contributing to psychiatric disease liability.

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INTRODUCTION

Impulsivity is a multi-faceted construct comprising several conceptually related phenotypes that reflect self-regulatory capacity. For example, one form of impulsivity is captured using delay discounting (DD), which measures a person's preference for smaller immediate rewards over larger future rewards [1]. Another class of well-established measures of impulsivity comes from self-reported personality questionnaires. For example, impulsive personality traits are typically assessed using the UPPS-P Impulsive Behavior Scales [2] or the Barratt Impulsiveness Scales (BIS; [3]). Unlike DD, impulsive personality can be further fractionated via subscales, measuring the tendency to act rashly under negative or positive emotions (Negative and Positive Urgency), the tendency to act without planning (Lack of Premeditation), the tendency to not complete tasks (Lack of Perseverance), and the tendency to enjoy intense sensation or experiences (Sensation Seeking).

These measures of impulsivity have been consistently linked to addiction [4–9] and other externalizing behaviors [10, 11], but with divergent patterns of association, suggesting multiple underlying dimensions [12, 13]. Further evidence of multi-dimensionality comes from genetic studies using a genomic structural equation modeling approach, confirming the five factors of the UPPS-P model and DD as genetically distinct phenotypes [14]. Much like alcohol and cigarette use [15, 16], impulsivity traits also present clear developmental trends [17, 18], suggesting these phenotypes comprise both stable trait-like features and more malleable state-like features.

To date, increasing progress has been made in establishing the genomic basis of impulsivity phenotypes measured cross-sectionally. The largest genome-wide association studies (GWASs) on DD [19] and impulsive personality traits (UPPS-P and BIS) together have identified 13 genome-wide associations [20, 21]. Though additional discovery of novel associations at single-SNP resolution will require even larger sample sizes, SNP-based heritability of ~5–15% has been reported for these traits [19–21]. These results suggest impulsivity traits are genomically

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complex and moderately heritable. However, unlike other complex traits (e.g., height), impulsivity could change during the life span due to changing environment and it is not clear whether genetic influences would remain stable. For example, the 'aging-out' process for substance use and impulsivity, has long been described in the literature [22], with supporting evidence of better genetic signal after aging out in twin studies [23]. Even though impulsivity has been shown to have moderate genetic continuity across adolescence in twin studies, its heritability increases as individuals age into adulthood [24, 25], suggesting ascertainment of genomic correlates may improve.

Indeed, GWAS under a longitudinal design broadly offers several benefits over traditional cross-sectional studies by providing increased statistical power and more precise estimations of genetic effects over time. Many human phenotypes can have both trait-like and state-like features, with the latter more likely to be influenced by situations, "aging out," or general maturation and reflected by changes over time. They also enable the detection of subtle genetic influences on the development and progression of traits and diseases, which might only manifest or vary over time. Additionally, longitudinal GWAS can better account for measurement error in general and temporal confounding (i.e, physical activity that fluctuate with the seasons). Finally, related findings suggest that a longitudinal design yields improved identification of genetic variants associated with alcohol use [26, 27] and body mass index [28, 29], which are phenotypically related to impulsivity [7, 30].

Increasingly, the importance of longitudinal trajectories has also been appreciated in the genetic investigation of clinical phenotypes that are known to fluctuate over time, such as blood pressure and fasting glucose [31]. However, genomic longitudinal analyses are not without challenges. For typical genomic analyses, large sample sizes are necessary to capture association signals at the SNP level while longitudinal studies typically are much smaller in size. This issue is further aggravated as the sample size tends to decrease over multiple follow-up visits. There are also analytical challenges, pertaining to modeling choices and handling of missing data. Latent growth curve (LGC) modeling is a statistical technique used within the structural equation modeling (SEM) framework to estimate growth trajectories and accounts for missingness in the data through full information maximum likelihood (FIML) estimation. The model identifies latent (unobserved) longitudinal features, such as intercepts (stable component) and slopes (rates of change) that define the shape of the growth curve whilst accounting for individual variability of growth. In contrast, the majority of longitudinal GWASs chose a mixed effect model over latent growth curve (LGC), and, while these are mathematically similar models, LGC is better suited for structured models with complex growth patterns, can tolerate missing covariates [32], removes measurement error in both predictors and outcomes, is statistically more powerful, and provides a meaningful interpretation of the changes in individual-level responses [33].

In this context, the current study had a number of aims. First, we characterized the longitudinal profile of several impulsivity phenotypes in the Avon Longitudinal Study of Parents and Children (ALSPAC) cohort, an ongoing longitudinal birth cohort study in the United Kingdom, examining both the longitudinal phenotypes and cross-sectional phenotypes over a four-year period. Specifically, we extracted the latent factor of intercept to capture both the stable trait-like feature of the cross-sectional phenotypes and state-like latent slope reflecting change over time,. Second, we examined whether a common set of genetic variants was responsible for the phenotypic stability over time by estimating the temporal genetic correlations between the same phenotype across time points. Third, building on the above aims, we estimated the SNP-based heritability of cross-sectional and longitudinal phenotypes to inform whether a longitudinal phenotype would produce a more robust heritability estimate. Specifically, we hypothesized that SNP-based heritability of the cross-sectional phenotypes that exhibited high genetic correlation over time would align with that of the longitudinal phenotypes, which were predicted to have greater statistical power for genomic discovery. This was examined using three association testing strategies: GWAS, which scans the entire genome agnostic as to the specific loci of interest; gene-based association, which focuses on genes previously linked to impulsivity; and polygenic risk scores (PRSs), which prioritizes traits that are genetic or phenotypic correlates of impulsivity, such as risk-taking [34]. The phenotypic characterization and association with PRSs were then validated in a replication sample that also included longitudinal impulsivity measures. The replication analyses were restricted to a subset of the primary analyses because the replication sample was considerably smaller and substantially underpowered for a number of analyses. Finally, to further address the potential utility of longitudinal phenotypes more generally, we conducted simulation studies to model and quantify the putative benefits.

METHODS

Study samples overview

The ALSPAC study is a UK-based birth cohort with roughly 15,000 children (Generation 1) and mothers (Generation 0) for whom genotype and phenotype data are available [35–37]. Specifically, pregnant women resident in Avon, UK with expected dates of delivery between 1st April 1991 and 31st December 1992 were invited to take part in the study; 20,248 pregnancies have been identified as being eligible and the initial number of pregnancies enrolled was 14,541. Measurements of impulsivity are available in three waves, at 24 (labeled by YPD), 26 (YPF), and 28 (YPH) years, in a subset of Generation 1 (G1) that completed the Monetary Choice Questionnaire (MCQ; [38]) and UPPS-P [2] at each visit. Study data were collected and managed using Research Electronic Data Capture (REDCap) tools hosted at the University of Bristol [39]. REDCap is a secure, web-based software platform designed to support data capture for research studies. The study website (http://www.bristol.ac.uk/alspac/researchers/our-data/) contains details of all the data that is available through a fully searchable data dictionary and variable search tool.

The present analyses focused on this subset of G1 genetic samples that completed the MCQ and UPPS-P in at least one of the YPD, YPF, and YPH waves to minimize samples removed for missingness. Each participant's response to the MCQ was used to calculate the discounting rates at three monetary reward magnitudes (small, medium, and large) otherwise referred to as small, medium and large *k*-values. MCQ data were further processed to ensure participants had answered the corresponding magnitude quality control items correctly. Three continuous DD phenotypes at large, medium and small reward magnitudes were calculated [40] and then log10-transformed. The final DD phenotype was obtained by averaging values over the three reward magnitudes, with higher values indicating a stronger preference for immediate rewards vs. delayed rewards. The five factors of UPPS-P were kept at their original scale, each with responses ranging from 4 to 16, where higher values denote higher impulsivity. These 6 phenotypes measured over three waves were used in all subsequent cross-sectional and longitudinal analyses (Supplementary Fig. 1).

For replication analyses, the PATH CANN study is an ongoing longitudinal cohort study of 1502 community adults based in Hamilton, Ontario, Canada, comprising repeated assessments of psychological processes (including impulsivity) and health behaviors. The overall sample has a wide age range, from 18 to 65, but was enriched for young adults that were 25 or younger (37%; 558 out of 1502). There are 12 waves of data completed between September 2018 and April 2023, which occurred in 6-month intervals except for two assessments that occurred in 3-month intervals during the height of the COVID-19 pandemic (between July 2020 and January 2021). Delay discounting was assessed using the 5-Trial Adjusting Delay Discounting Tasks of \$100 and \$1000 [41] magnitudes at all time points. The 5-Trial DD produced *k*-values for those who correctly answered the quality control item for the magnitude (larger versus smaller amount with no delay) and were subsequently log10-transformed. The final DD phenotype was obtained by averaging values for the two reward magnitudes. Impulsive personality traits were assessed using the 5 UPPS-P subscales [2] with values between 4 to 16. Participants were included if they had at least two or more waves of data that also passed quality control indicators (96%; $n = 1,449$; mean waves $= 12.1$; median waves $= 13$).

Genetic data

Genotyped data were available for 8932 G1 samples. Basic quality control steps (minor allele frequency; MAF > 0.01, sample and genotype missing rate < 0.05, and Hardy-Weinberg disequilibrium p-value $> 5 \times 10^{-7}$) were applied prior to imputation using the TOPMed imputation Server [42]. After imputation, standard quality control filters were used to retain unique bi-allelic SNPs with INFO score > 0.7, MAF > 0.01, sample missing rate < 0.05, genotype missing rate < 0.05, and Hardy-Weinberg disequilibrium p-value $> 5 \times 10^{-7}$. Genetic ancestry was confirmed using the 1000 Genomes Project [43] as the reference panel (Supplementary Fig. 2). For a step-by-step sample/SNP inclusion and exclusion, see Supplementary Fig. 3. The final analytical data contained 7975 unrelated, genetically confirmed European samples on 9,248,678 autosomal and 252,917 X chromosome SNPs. After merging with the impulsivity phenotypes, the data contained roughly 3000 individuals at each wave. A summary of the final analytical dataset for a cross-sectional design with quality-controlled genetic data at YPD, YPF, and YPH can be found in Supplementary Table 1. Genotype data quality control for PATH CANN had been previously described [44]. The final European ancestry sample containing both quality controlled genetic and phenotypic data for PATH CANN was 1019. Thus, we only reported a subset of the genetic results from PATH CANN due to concern for sample size and power as heritability and co-heritability methods usually require at least 3000 samples to obtain reliable estimates [45].

Longitudinal phenotypes using a latent growth curve model

To establish the basis of a longitudinal modeling approach, we first examined the phenotypic correlation pattern and trends for DD and the five factors of UPPS-P over the three waves. We then applied a LGC model to each phenotype over all time points, in which we tested (1) a no-growth model and (2) a linear growth model. Age at the first time point was added as a time-invariant covariate in the models, meaning that the growth factors were regressed on age. For PATH CANN, age at the first time point as calculated from date of birth was used to control for the varying ages in the cohort. For ALSPAC, Age at YPD was added as a time-invariant covariate in the models as there was some variability in reported age due to lengthier data collection periods. Missing values were imputed using full information maximum likelihood implemented in the R package "*lavaan*" [46]. The LGC model yielded 6 intercept phenotypes for a no-growth model, or 6 intercept and 6 slope phenotypes for a linear growth model. These were interchangeably referred to as LGC phenotypes or longitudinal phenotypes. We also included the average, referred to as the Mean Model, as an alternative longitudinal approach in the simulation studies, heritability estimation, and the PRS association testing to demonstrate the statistical advantage of modeling higher-order changes. The Mean Model imputed missing values by the mean measurements from the available waves and thus had the same final sample size as the LGC approach.

SNP heritability and co-heritability

For cross-sectional and longitudinal phenotypes, the SNP-based heritability and temporal genetic correlation were estimated using LDscore regression [47] based on common SNPs present in HapMap3. For cross-sectional phenotypes, we only examined the temporal genetic correlation between those with estimated heritability greater than 0.05 to avoid the issue of model convergence. Given the sample size in ALSPAC (~ 3000), we expected to have $> 80\%$ power to detect a significant heritability of 0.3 and roughly 60% power to produce a heritability of 0.2 using LD regression methods at a nominal p-value threshold < 0.05 (https://nealelab.github.io/UKBB_ldsc/confidence.html#minimum_sample_size).

GWASs and gene-based analyses

A total of 18 cross-sectional GWASs were conducted on DD and the 5 UPPS-P subscales over the three waves. For autosomal SNPs, a linear regression model was used to test the association between the genotype of each SNP, coded additively as the number of minor alleles, with the phenotype while adjusting for genetic sex, age at each wave, and the first 10 genetic PCs. X chromosome association testing was performed using a robust method designed for X chromosome [48]. For the derived longitudinal phenotypes, GWAS was performed on the 6 latent intercepts. The genetic association was evaluated using the same model except for age at each wave. All autosomal analyses were conducted in PLINK2 [49], while the X chromosome analyses were performed using Statistical Software R version 4.0.5 [50].

We used MAGMA v1.08 [51] to perform a gene-based association analysis for longitudinal intercepts by focusing on 28 genes (Supplementary Table 2) that had either been mapped to SNPs previously identified at genome-wide significance with any impulsivity traits (DD, UPPS-P, and BIS-11) or because their predicted gene expression had been linked to these traits. For each prioritized gene, we also tested the association between their predicted gene expression derived from transcriptome-wide association results from GTEx ver. 8 [52] and LGC impulsivity phenotypes in ALSPAC.

Polygenic risk scores analysis

Summary statistics for DD and UPPS-P subscales [19, 20], externalizing [53], general risk tolerance [54], neuroticism [55], general addiction factor [56], problematic alcohol use [57], alcohol consumption, and smoking phenotypes [58] were obtained from published GWASs (Supplementary Table 3). We matched SNPs and their reference alleles between the summary statistics data and our ALSPAC genetic samples. Lassosum [59] was used to construct PRSs, whereby each PRS was validated using the target phenotype in ALSPAC that matched most closely to that in the original GWAS. If a target phenotype was not available, a pseudo-validation procedure was used instead [59]. Each derived PRS was then tested for association with cross-sectional and longitudinal impulsivity intercept phenotypes after adjusting for genetic sex and the first 10 genetic PCs. For PATH CANN, only PRSs for impulsivity measures (DD and UPPS-P subscales) were derived and tested with the corresponding longitudinal phenotype, while additionally adjusting for income, age, and genotyping array batch. We reported the adjusted R^2 calculated as the difference between the full model (PRS and all covariates) and the reduced model (only covariates). All statistical analyses were conducted in R version 4.0.5.

Simulation studies

As the premise of using latent phenotypes to improve genetic signals is based on reducing random measurement noise over multiple time points, we further tested this hypothesis using a simulation study where each phenotype y was partitioned to have a common genetic component, an autocorrelated environmental component, and a random noise that is independent of time and other components. Let t denote an integer time point that starts at $t = 1$. The phenotype for the ith individual at the tth time point can be expressed as:

$$y_{it} = g_i + f_{it} + \varepsilon_{it}$$

where the genetic component $g_i \sim N(0, \sigma_g^2)$ was assumed to be sampled from a normal distribution with variance σ_g^2, the random measurement error $\varepsilon_{it} \sim N(0, \sigma_\varepsilon^2)$ was normal with variance σ_ε^2, and an environmental component $f_i \sim N(\mathbf{0}, \mathbf{\Sigma_f})$ that was assumed to be multivariate normal with covariance $\mathbf{\Sigma_f}$. The covariance structure is specified by the amount of correlation between time points with $\mathbf{\Sigma_f}(t, t') = \sigma_f^2 \phi^{t-t'}$, where $0 < \phi < 1$. For the simulation study, we fixed $\sigma_g^2 = 0.2$, $\sigma_f^2 = 0.8$, and $\sigma_\varepsilon^2 = 0.3$, such that the theoretical heritability and the theoretical upper bound of heritability estimate, defined as

$$\frac{\sigma_g^2}{\sigma_g^2 + \sigma_f^2}$$

and

$$\frac{\sigma_g^2}{\sigma_g^2 + \sigma_f^2 + \sigma_\varepsilon^2}$$

were 0.2 and 0.154, respectively. We considered 12 scenarios by taking combinations of 3, 5, and 10 time points, and sample sizes equal to 2500, 5000, 10,000, and 20,000. A total of 1000 simulation runs were performed for each scenario. Across simulations, the genetic component and the environmental components were simulated only once but the error terms were sampled at each simulation run.

RESULTS

Longitudinal impulsivity phenotypes

For DD and UPPS-P subscales, 3969 participants completed at least one wave and 3856 had valid data for at least one time point

(Supplementary Table 1). Among those with data for at least one wave, roughly 76% had data for 2 waves and 50% had data for all three waves. The alluvial plot showed the proportion of missingness based on those who would be included in the LGC model (100% complete as LGC uses FIML that requires 1+ waves) vs. each wave (Supplementary Fig. 4). Using DD as an example, we performed a sensitivity analysis showing that individuals that were more impulsive (-2.07 ± 0.46 vs. -2.17 ± 0.48; two-sample t-test $p = 6.6 \times 10^{-12}$) were more likely to not participate in all three waves.

For all impulsivity indices, the linear growth was a better model judging by the model fit indices (Table 1), although, the slope estimate was not statistically significant for Negative Urgency, and the size of the remaining slope factor varied from trait to trait (Table 1). Most traits exhibited a significant small effect size downward trend in the ALSPAC sample (Supplementary Fig. 5), with Sensation Seeking (slope $= -0.27$; $p < 2 \times 10^{-16}$; Table 1) and DD (slope $= -0.051$; $p = 4.4 \times 10^{-16}$) having the most robust decrease over time. In comparison, we observed similar downward trends for Sensation Seeking and DD in PATH CANN (Table 1), but no change for Negative Urgency, Positive Urgency and Lack of Premeditation, and only a significant small increase in Lack of Perseverance. Since there was minimal mean difference in the intercept phenotype estimated under the no-growth vs. linear growth model, for the remaining analyses, the intercept phenotypes for DD and all UPPS subscales estimated under the linear growth were used based on improved model fit.

In terms of correlation among impulsivity measures, the LGC intercept phenotypes showed better structural consistency in comparison to their cross-sectional counterparts (Fig. 1A). Specifically, Lack of Perseverance correlated better with Lack of Premeditation using the LGC intercept phenotype ($r = 0.43$) than cross-sectional measures ($r = 0.28-0.42$), similarly for Positive and Negative Urgency ($r = 0.61$ vs. $r = 0.40-0.57$). Further, the correlation structure among all LGC intercept phenotypes was highly consistent between ALSPAC and PATH CANN samples (Supplementary Fig. 6).

SNP-based heritability and genetic correlation

For DD, temporal genetic correlation was high between consecutive waves ($r_g = 0.77$, 0.53, and 1.0; $p = 0.076$–0.4; Fig. 1B; Supplementary Table 4), but slightly lower between the first and the third wave ($r_g = 0.53$). The patterns of these temporal genetic correlations were similar to those observed phenotypically ($r = 0.57-0.66$). On the other hand, Sensation Seeking exhibited much stronger temporal correlations genetically ($r_g = 1$; $ps < 0.0000003$; Supplementary Table 4) as compared to that phenotypically ($r = 0.76-0.78$; Fig. 1B). We also observed high genetic correlation between Negative Urgency and Lack of Premeditation at YPD ($r_g = 0.93$; $p = 0.02$) and between Lack of Perseverance of Lack of Premeditation at YPF ($r_g = 0.8$; $p = 0.04$). But here we only focused on genetic correlation estimates for traits that exhibited a significantly non-zero heritability due to concern for model convergence and power.

The cross-sectional heritability estimates of DD (SNP-based $h^2 = 0.12$–0.15; $p = 0.19$, 0.22, 0.13) and Sensation Seeking (SNP-based $h^2 = 0.31$–0.40; $p = 0.0066$, 0.024, 0.022) varied by wave (Fig. 2; Supplementary Table 5), and these were the only two impulsivity measures with SNP-based heritability consistently above 0.05 across waves. In contrast, heritability estimates for the remaining UPPS-P subscales varied and exhibited more variability over time. In fact, heritability at the first (YPD) and the last wave (YPH) had values below zero (Supplementary Table 5), reflecting non-convergence. As a result, co-heritability for those with negative or very small estimated heritability did not converge (Supplementary Table 4).

In contrast, the heritability of the longitudinal intercept ranged from $0.07-0.35$, showing generally improved heritability as compared to their cross-sectional counterparts and the Mean Model (Supplementary Table 5). In particular, the DD intercept produced a statistically significant heritability estimate with a smaller variance ($h^2 = 0.22$, s.e. $= 0.12$, $p = 0.03$; Fig. 2), whereas no cross-sectional estimates reached nominal significance. The heritability estimate of the Sensation Seeking LGC intercept phenotype aligned better with values derived under the cross-sectional design ($h^2 = 0.35$, s.e. $= 0.11$; Fig. 2) and afforded stronger statistical evidence than the cross-sectional measures ($p = 0.0007$ vs. $p = 0.025$–0.0066). Overall, two of the six impulsivity indicators had significant SNP-heritability using the LGC intercept vs. one of the six when examining their cross-sectional counterparts (Supplementary Table 5). For the remaining four indicators, the LGC intercept produced positive heritability for Negative Urgency ($h^2 = 0.15$, s.e. $= 0.13$, $p = 0.12$), Positive Urgency ($h^2 = 0.09$, s.e. $= 0.10$, $p = 0.19$), Lack of Premeditation ($h^2 = 0.17$, s.e. $= 0.13$, $p = 0.10$), and Lack of Perseverance ($h^2 = 0.07$, s.e. $= 0.11$, $p = 0.27$). Using the longitudinal intercepts, we produced a genetic correlation matrix for DD and UPPS-P subscales (Fig. 1C), where Negative Urgency was positively correlated with Positive Urgency ($r_g = 0.85$, $p = 0.02$; Supplementary Table 6).

Cross-sectional and longitudinal GWAS results

At genome-wide significance ($p < 5 \times 10^{-8}$), there were 3 significant associations from individual cross-sectional GWASs (Supplementary Table 7): Sensation Seeking at YPF (24-year) was associated with rs221655 on chromosome 6 in the *BVES* gene, and Positive Urgency at YPH (26-year) was associated with rs141363425 and rs117206739 on chromosome 18. In contrast, we found only 1 genome-wide significant association from the 6 LGC GWASs (Supplementary Table 8), where rs11984107 in *TMEM209* on chromosome 7 was associated with Lack of Perseverance. In the Manhattan plots of the 6 LGC GWASs (Supplementary Figs. 7–12), we highlight SNPs within 5Kbp distance up- and downstream of the 28 genes if their corresponding p-values were below 0.005. Among them, *CADM2*, *CACNA2D1*, *CADPS2*, *ELOVL7*, *MDM1*, *RAP1B*, *TMEM161B*, *MEF2C*, *TCF4*, *SPHKAP*, *PID1*, *XKR6*, *NCAM1*, and *GPM6B* genes were significant ($p < 0.005$) with multiple LGC impulsivity phenotypes (Supplementary Table 2). Of particular note, *CADM2* and *CACNA2D1* were implicated with all impulsivity indices except Positive Urgency.

Gene-based analyses

The gene-based analysis of LGC intercepts confirmed the association between the *CACNA2D1* gene and Lack of Premeditation after false discovery rate (FDR) adjustment at an alpha-level of 0.05 ($p = 0.0013$, $q = 0.03$; Supplementary Table 9). At a more relaxed nominal p-value threshold of 0.05, 4 out of the 28 genes were suggestively associated with Sensation Seeking, 3 for Lack of Perseverance, and 2 each for Positive and Negative Urgencies. There was only one suggestive association for DD ($p = 0.017$), implicating the *NCAM1* gene on chromosome 11. There was complementary evidence from the prediXcan results for the association between the *CACNA2D1* gene and Lack of Premeditation combining the associations from all tissues ($p = 0.00015$, $q = 0.043$; Supplementary Table 10). Further, the genetically predicted expression of *XKR6* was also significantly associated with Negative Urgency across tissues ($p = 0.00015$, $q = 0.0035$; Supplementary Table 10). Though not statistically significant after adjusting for multiple testing, *CLDN23* and *CACNA2D1* were suggestively associated with multiple impulsivity indices ($p = 0.0094$–0.041 and $p = 0.015$–0.05, respectively) using MAGMA and implicated in prediXcan results (Supplementary Tables 9–10).

Polygenic risk score analyses

Out of the 17 PRSs constructed, 16 demonstrated reasonable in-sample properties (validation association $p = 0.015-1.4 \times 10^{-28}$,

Table 1. Latent growth curve model fit indices and parameter estimates for ALSPAC and PATH CANN samples.

Trait		Negative urgency		Positive urgency		Sensation seeking		Lack of premeditation		Lack of perseverance		Delay discounting	
	LGC model	**No growth**	**Linear growth**	**No growth**	**Linear growth**	**No growth**	**Linear growth**	**No growth**	**Linear growth**	**No growth**	**Linear growth**	**No growth**	**Linear growth**
ALSPAC ($n = 3860$)	AIC	40745.56	40734.89	37208.21	37154.63	38053.36	37867.05	35036.99	35000.68	34636.85	34604.79	15545.30	15471.69
	BIC	40795.60	40803.69	37258.25	37223.43	38103.40	37935.85	35087.01	35069.47	34686.88	34673.58	15595.36	15540.53
	RMSEA	0.04	0.04	0.04	0.02	0.09	0.06	0.03	0.00	0.03	0.01	0.05	0.03
	CFI	0.98	0.99	0.97	1.00	0.94	0.98	0.98	1.00	0.98	1.00	0.96	0.99
	TLI	0.98	0.98	0.97	0.99	0.93	0.97	0.98	1.00	0.98	1.00	0.96	0.99
	Estimates (Intercept)	**8.66**	**8.65**	**6.66**	**6.79**	**10.46**	**10.77**	**7.25**	**7.34**	**7.28**	**7.34**	**−2.18**	**−2.12**
	P-value (Intercept)	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**
	Estimates (slope)		0.022		**−0.10**		**−0.27**		**−0.086**		**−0.054**		**−0.051**
	P-value (slope)		0.38		**0.0000005**		**<0.00000001**		**0.000003**		**0.0035**		**<0.00000001**
PATH CANN ($n = 1449$)	AIC	70212.60	65502.64	61710.07	59713.65	59071.03	22019.98	69452.36	64789.63	60677.65	59236.07	58633.89	21144.50
	BIC	70297.05	65587.09	61794.51	59798.09	59155.48	22104.42	69557.92	64895.19	60783.20	59341.63	58739.45	21250.06
	RMSEA	0.09	0.09	0.10	0.07	0.07	0.10	0.06	0.06	0.06	0.04	0.05	0.07
	CFI	0.93	0.94	0.95	0.96	0.95	0.94	0.97	0.97	0.98	0.99	0.98	0.97
	TLI	0.94	0.94	0.95	0.96	0.96	0.94	0.98	0.98	0.98	0.99	0.98	0.98
	Estimates (Intercept)	**9.41**	**9.50**	**8.23**	**8.21**	**11.65**	**11.88**	**7.22**	**7.17**	**7.62**	**7.47**	**−2.20**	**−2.12**
	P-value (Intercept)	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**	**<0.00000001**
	Estimates (slope)		−0.0092		0.0012		**−0.024**		0.0057		**0.016**		**−0.0070**
	P-value (slope)		0.26		0.87		**0.0002**		0.33		**0.0063**		**0.0020**

The Bold font indicates estimates reached statistical significance ($p < 0.05$).

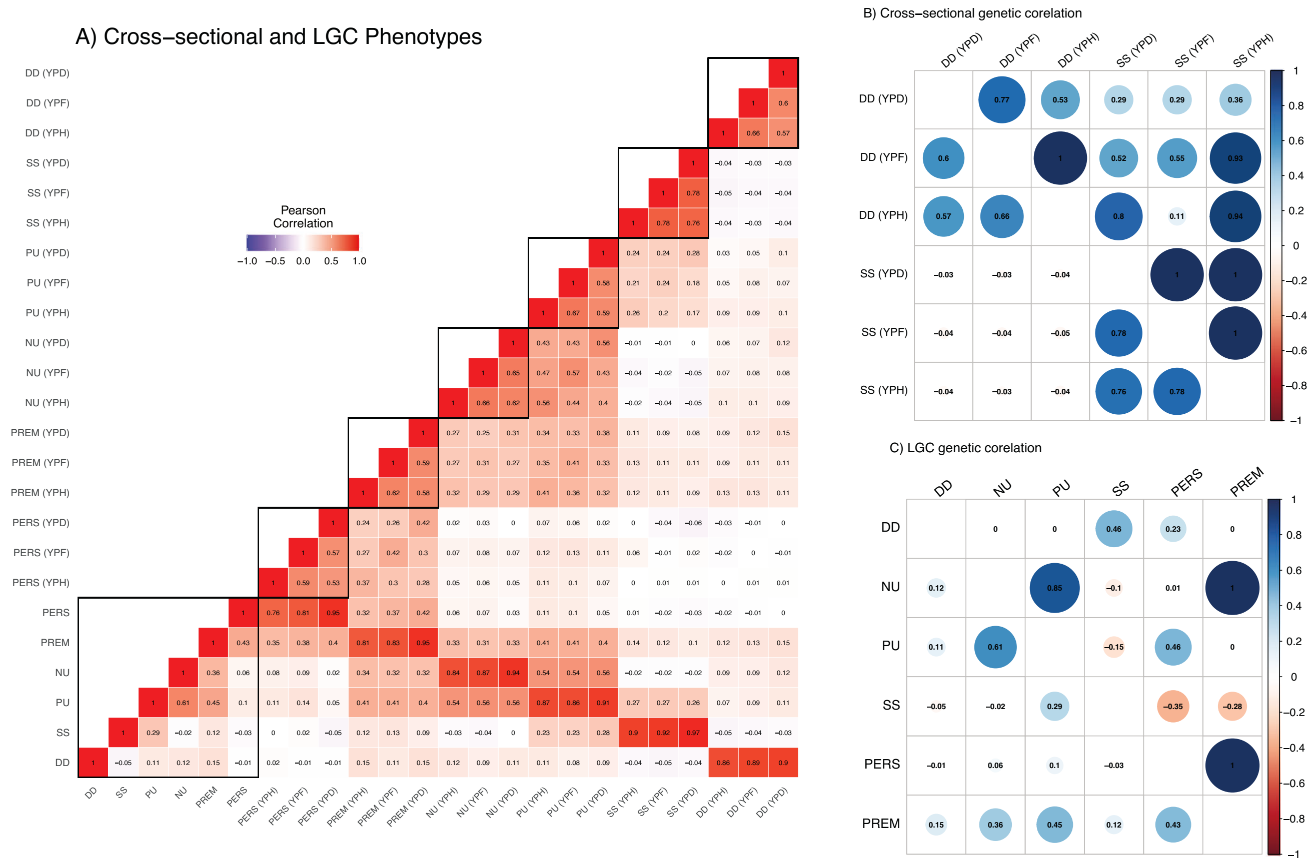

Fig. 1 Heatmap of phenotypic and genetic correlation of cross-sectional and longitudinal impulsivity intercepts. Heatmap in Panel **A** displays the correlation matrix between cross-sectional and longitudinal impulsivity measures. Each cell represents the correlation coefficient between the variables on the corresponding *x* and *y* axes, ranging from −1 (strong negative correlation, shown in blue) to +1 (strong positive correlation, shown in red). Cells with a correlation close to 0 are colored in neutral (white), indicating no correlation. The color intensity increases with the strength of the relationship. The heatmap in Panel **B** shows the phenotypic and genetic correlation matrices between cross-sectional DD and Sensation Seeking. The phenotypic correlations were labeled in the lower triangular area of the matrix while the genetic correlation was in the upper triangular area. Both the color intensity and diameter of the circle increases with the strength of the relationship. The heatmap in Panel **C** shows the phenotypic and genetic correlation matrices between longitudinal impulsivity intercepts. The phenotypic correlations were labeled in the lower triangular area of the matrix while the genetic correlation was in the upper triangular area.

adjusted $R^2 = 0.27\%–12.35\%$; Supplementary Table 11) and were retained for association analysis. The exception was the PRS of Lack of Premeditation, which was not significantly associated with the target phenotype in ALSPAC after FDR correction ($p = 0.048$, $q = 0.055$) nor with any other impulsivity measures (Supplementary Table 11), and was thus removed. Between cross-sectional and longitudinal impulsivity phenotypes, the general patterns of association with PRSs were similar (Fig. 3), but the longitudinal phenotypes generally had larger effect sizes and smaller *p*-values than their cross-sectional counterparts (Supplementary Table 12). As a result, there were more significant PRS-impulsivity associations with longitudinal and the mean phenotype than cross-sectional phenotypes after FDR correction (47 and 46 vs. 35–42; Supplementary Fig. 13), boosting signal by 9.5%–34%.

Among the 16 validated PRSs, 5 were specific to impulsivity indicators of DD and UPPS-P scales, while 6 were general measures of externalizing behaviors or personalities, and the remaining 5 captured alcohol and tobacco use. First, all 5 impulsivity PRSs were significantly associated with their respective longitudinal intercept or cross-sectional phenotype at least one of the waves (Supplementary Fig. 14; Supplementary Table 13), explaining 0.18−0.67% of the LGC intercept phenotypic variance. Secondly, all impulsivity indicators – cross-sectional or longitudinal – were associated with at least one of the behavioral PRSs, but showed distinct patterns in their associations with these PRSs (Supplementary Fig. 15; Supplementary Table 12). Remarkably, Positive Urgency was consistently associated with all 6 PRSs (Supplementary Table 12), while Negative Urgency was associated with all but the Adventurousness PRS, with a similar pattern but weaker association strengths with respect to the remaining PRSs. As expected, Negative Urgency's association with the Neuroticism PRS was elevated (adjusted $R^2 = 1.40\%$) as compared to that of Positive Urgency (adjusted $R^2 = 0.30\%$), and both in the negative direction. Similarly, Sensation Seeking was positively associated with the Adventurousness PRS (adjusted $R^2 = 3.88\%$), the strongest signal across all associations tested, then the General Risk Tolerance PRS (adjusted $R^2 = 1.58\%$), the Externalizing PRS (adjusted $R^2 = 0.57\%$), and negatively with the Neuroticism PRS (adjusted $R^2 = 0.30\%$). Between the impulsivity indicators, Lack of Premeditation showed the strongest association with the Addiction-rf PRS (adjusted $R^2 = 0.47\%$), but not with the Neuroticism PRS. On the other hand, DD was associated with PRSs of Externalizing, General Risk Tolerance, ADHD, and Neuroticism, but not with the Addiction-rf and Adventurousness PRSs.

Moreover, the impulsivity indicators showed specific association patterns with substance use PRSs according to the severity and stages of use. Both alcohol PRSs were strongly associated with four of the UPPS-P scales (excluding Lack of Perseverance) but not with DD (Supplementary Fig. 16; Supplementary Table 12). In general, we observed stronger associations with the PAU PRS

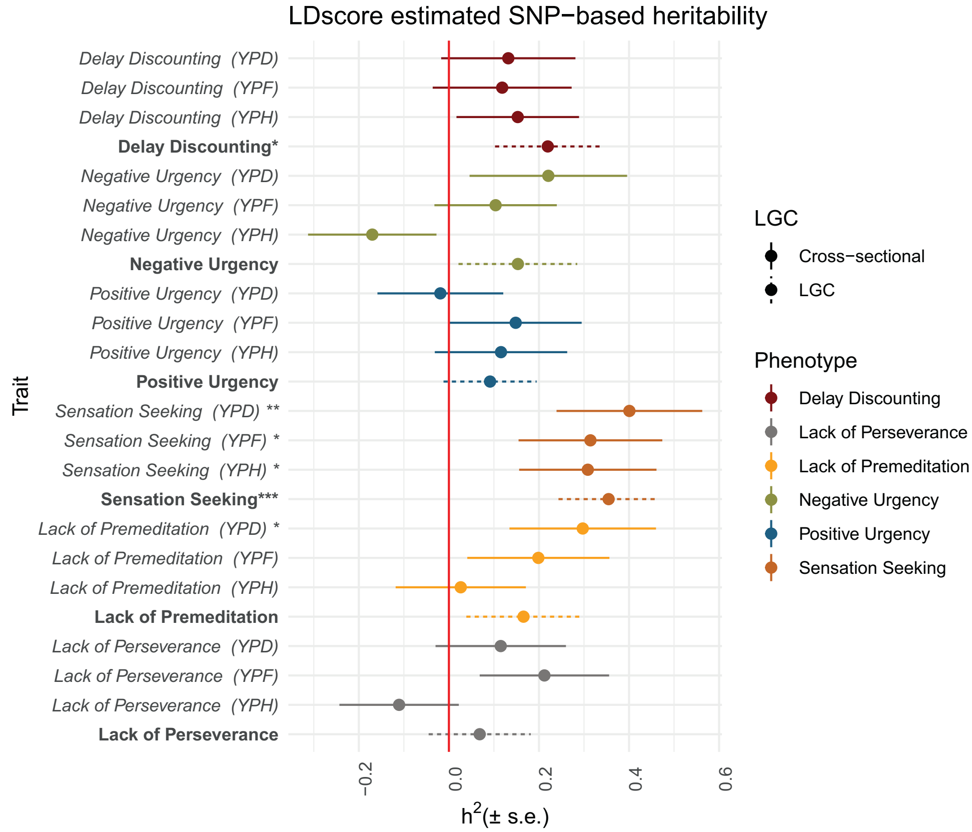

Fig. 2 SNP-based heritability of cross-sectional and longitudinal impulsivity intercept phenotypes estimated using LDscore. The *x*-axis represents the estimated heritability values (+/− s.e.) and the y-axis shows the impulsivity phenotypes at each wave (YPD, YPF, or YPH) in solid lines and the longitudinal intercept phenotype in dashed lines. A single asterisk indicates a phenotype has a statistically significant heritability estimate at a nominal threshold of $p < 0.05$, two asterisks correspond to $p < 0.01$, and three asterisks correspond to $p < 0.001$.

(adjusted $R^2 = 0.49-0.68\%$) than the Drinks per Week PRS (adjusted $R^2 = 0.22-0.32\%$), with the exception of Sensation Seeking where its association was stronger with the Drinks per Week PRS (adjusted $R^2 = 0.22\%$; Supplementary Table 12) than the PAU PRS (adjusted $R^2 = 0.17\%$). On the other hand, DD and four of the UPPS-P scales (excluding Lack of Perseverance) were implicated with smoking PRSs with distinct patterns across the stages of smoking. We observed positive associations between DD, Positive and Negative Urgencies, and Lack of Premeditation with the Smoking Initiation PRS (adjusted $R^2 = 0.24-0.6\%$) with Negative Urgency showing the strongest effect (adjusted $R^2 = 0.60\%$; Supplementary Table 12). However, only DD was associated with the Smoking Cessation PRS (adjusted $R^2 = 0.20\%$; $p = 0.002$) and there was no significant association between any impulsivity measure and the Cigarette per Day PRS.

Polygenic risk score analyses in replication samples

In the replication cohort with 13 waves but a smaller genomic sample ($n = 1,019$), we observed significant evidence of association ($ps < 0.05$) between the impulsivity PRSs with their corresponding longitudinal intercept phenotypes for 4 out of the 6 impulsivity measures (Supplementary Table 13). Notably, the association strengths were highly consistent with those observed in ALSPAC for DD (adjusted $R^2 = 0.45$ vs. 0.67%), Negative Urgency (adjusted $R^2 = 0.44$ vs. 0.61%), Sensation Seeking (adjusted $R^2 = 0.59$ vs. 0.36%; Supplementary Table 13). Among the four impulsivity measures significantly associated with their respective PRSs, the LGC intercept phenotypes generally had smaller standard errors and, in turn, were more statistically significant as compared to the cross-sectional phenotypes (Supplementary Fig. 17).

Simulation studies

Under the model specified, the LGC phenotype had the best heritability estimates in comparison to cross-sectional measurements across scenarios (Supplementary Fig. 18). The improved performance was marked by the distance to the true heritability (red line) and the standard error (proportional to the width of the box). As the sample size increased, heritability estimates of both LGC and cross-sectional phenotype converged to the true values marked by the red and black lines, respectively. However, we found that ignoring the time aspect by taking the average can lead to over-estimation when the number of longitudinal measurements was large (e.g., 10 waves). The magnitude of this bias also depended on the true autocorrelation between the non-genetic variance component (Supplementary Fig. 18). In contrast, we observed generally better heritability estimates using the LGC phenotype with an increasing number of time points, owing to the model's ability to further "average out" the noise component.

DISCUSSION

In this work, we constructed longitudinal impulsivity phenotypes in the ALSPAC cohort by characterizing the stable trait-like component using latent intercepts and the consistently malleable component using latent slopes. Subsequently, correlations among these latent LGC phenotypes were assessed in contrast to longitudinal phenotypes, offering insights into the longitudinal trajectory of self-regulatory phenotypes in a broader demographic context. Following this, we evaluated SNP-based heritability and genetic correlation, separately for cross-sectional and the derived LGC intercept phenotypes. The heritability and genetic correlation estimates for the LGC intercept phenotypes were more consistent

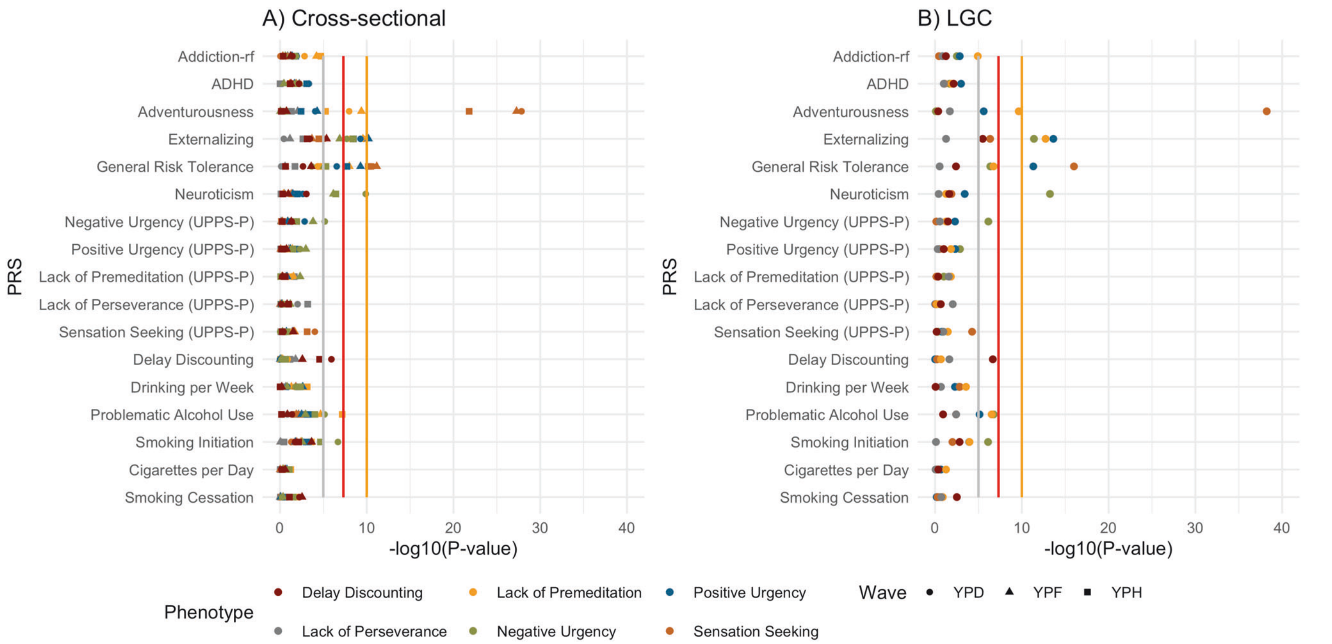

Fig. 3 PRSs association with cross-sectional and longitudinal impulsivity intercept phenotypes. In panels **A** and **B**, the *x*-axis represents the -log10 of association *p*-value between each PRS and the impulsivity phenotype, and the *y*-axis shows the corresponding PRS. Panel **A** summarizes results for cross-sectional phenotypes at each wave with colored dots representing the impulsivity phenotype (DD or one of the UPPS-P subscales) and the shape representing the wave (YPD, YPF, or YPH). Panel **B** summarizes the results for longitudinal phenotype with each colored dots representing the latent intercept for that impulsivity phenotype (DD or one of the UPPS-P subscales). The gray, red, and orange vertical lines indicate -log10 *p*-value thresholds of $p < 1 \times 10^{-5}$, $p < 5 \times 10^{-8}$, and $p < 1 \times 10^{-10}$, respectively.

with previous studies [19–21] using 6 to 35 times the sample size of ALSPAC. Our PRS association study also supported the use of LGC intercept phenotypes, which were further validated through impulsivity PRSs and their longitudinal intercepts in an independent longitudinal study. A replication sample provided converging evidence on the phenotypic structure of the impulsivity indicators and associations with within-trait PRSs. Results from the simulation studies also confirmed that longitudinal phenotypes are robust and less noisy instruments for genomic analyses and have the potential to improve signal discovery.

The benefit of a longitudinal modeling strategy for self-regulatory is two-fold, providing a robust phenotypic and genotypic characterization of the underlying phenotype. The phenotypic characterization of impulsivity using the LGC phenotypes was consistent with epidemiological observation of aging out, i.e. decreased impulsivity with age. As compared to the simpler Mean Model, LGC improves measurement resolution of the changes in the critical young adult period that is the mid-to-late twenties, a significant phase in many people's lives due to several developmental milestones and societal expectations that often occur during this time. For example, we found meaningful changes in Sensation Seeking and DD, with evidence of an average young adult becoming less impulsive as they age that is consistent with literature [60, 61]. This is putatively because LGC permits partitioning of changes into estimates of stable intercept component, a stable change component (increases or decrease), and error. In addition, more frequent observations can lead to increased precision in estimates of change, as there is less opportunity for unobserved influences to affect the measurements between time points. Finally, it increases the precision of measurement and allows for a refined model specification, accounting for autocorrelation, time-varying confounders, and is robust to violations of missing at random [62], which we had shown to be the case for impulsivity phenotypes. From a genomic perspective, the most remarkable gain was the improvement of SNP-based heritability for impulsivity. By contrasting the cross-sectional and longitudinal heritability estimates, we interpreted the results to suggest that Sensation Seeking has the least amount of measurement noise and the most stable genetic component over time. For DD, the genetic component also appeared to be stable over the three waves, but the almost 50% gain in SNP-based heritability pointed to a non-negligible amount of noise variance.

In terms of signal discovery, though we were underpowered to identify individual SNPs associated with impulsivity, the a priori gene-based analyses confirmed previously reported SNP associations at the gene level between SNPs in *XKR6* with Negative Urgency and three other genes (*CADM2*, *CLDN23*, and *CACNA2D1)* that were pleiotropic with several related traits [20, 21]. Notably, *CADM2* had been linked to a wide range of cognitive, risk taking, alcohol use, and food preference traits [63, 64], while *CLDN23* was robustly associated with neuroticism [65] and general risk tolerance [54]. Collectively, the gene-based analyses confirmed the role of *CADM2*, *CACNA2D1*, *CLDN23*, and *XKR6* in influencing the genetics of impulsivity.

Further evidence also comes from the PRS association results, in which we have indirect support for the genetic correlation between Positive and Negative Urgencies (Supplementary Table 6; Supplementary Table 12) and the genetic overlap between Negative Urgency, neuroticism, and general risk tolerance. More importantly, the list of self-regulation PRSs revealed distinct patterns of association with respect to different impulsivity measures, suggesting multiple underlying dimensions with both overlapping and unique genetic components. For example, while all impulsivity measures were positively associated with the externalizing PRSs, a subset of impulsivity measures, including DD, Positive and Negative Urgencies were negatively associated with the neuroticism PRS. Further, DD was significantly associated with the PRSs of smoking initiation and cessation, but not with the PRSs of cigarette quantity nor alcohol-related traits, which suggested that impulsivity was associated with tobacco onset and tobacco use disorder recovery but not level of tobacco use, which is putatively driven by genes involved in nicotine pharmacokinetics and pharmacodynamics [66, 67]. In addition to the aforementioned confirmatory findings from our gene-based analyses and GWASs, these results also lent further support for the role of *NCAM1* in DD, which has previously been implicated [68], and thus providing a hypothesis for one genetic mechanism in the established negative prognostic relationship between impulsive DD and smoking cessation [69].

However, our study was not without limitations. First, while it seemed unconventional to estimate the temporal genetic correlation, that is, the genetic correlation of the same trait measured on the same individuals across different time points, these estimates represent an upper bound for the stable component of genetic contribution. In other words, large genetic correlation estimates are the first line of evidence supporting a genomic investigation of longitudinal phenotypes. In that regard, we were unable to examine the temporal genetic correlation of all impulsivity traits due to four of the UPPS subscales being underpowered in certain waves (Positive Urgency at YPD and the remaining ones at YPH). Second, though the stable trait-like feature was consistent between the two longitudinal models, there appeared to be non-linearity over the three waves for Negative Urgency, while other impulsivity indicators did not significantly deviate from their linear trends (Supplementary Fig. 5). Notably, the last wave (YPH) of the ALSPAC data was collected in 2020 and coincided with COVID-19, and there has been research work linking COVID-related stress to increased impulsivity [70]. In contrast to other impulsivity indicators, the divergence in Negative Urgency's longitudinal trend may be attributable to the increased stress and hardship resulted from COVID-19 that triggered negative emotions. The inconsistent trend between ALSPAC and PATH CANN with respect to Lack of Perseverance also illustrated the potential distortions in longitudinal trends that arose when impulsivity measurements were gathered across a broad period of human development, with only robust and replicable "aging out" evidence for Sensation Seeking and DD. Finally, while several previous findings were confirmed in our gene-based analyses, we were unable to replicate individual SNPs at genome-wide significance due to the sample size. Indeed, the relatively small sample size was a constraint and a consideration across our findings, albeit one that is mitigated somewhat by the relatively robust findings that emerged within that context. Acknowledging these limitations, a strength that should be noted is the unique design comprising three longitudinal observations over four years of several thousand approximately age-matched individuals of common ancestry. While larger GWASs of impulsivity utilizing cross-sectional designs are essential, a longitudinally informed genetic strategy in a smaller study can be complementary since it allows us to model both the stable and the time-varying trend under genetic influence. In the absence of fluctuation due to genetic factors or gene-time interactions, aggregating repeated measures can better capture the genetic signals by reducing temporal noise such that the upper bound of the estimated heritability can approach its theoretical value.

To conclude, we show that the constructed longitudinal impulsivity phenotype better characterized underlying traits in terms of phenotypic structure and genetic contributions. These results further extend the relatively small body of literature on the genomic basis of impulsivity phenotypes and their genetic correlates via PRSs, further confirming a substantive genetic contribution to impulsivity. Both empirical and simulation results demonstrated the potential of longitudinal phenotyping to boost genomic discovery for self-regulatory phenotypes and their linkage to other traits and psychiatric disorders. Ultimately, longitudinal studies with larger sample size and increased number of time points are needed to pinpoint specific genetic factors that contribute to the development of self-regulatory phenotypes.

DATA AVAILABILITY

Genotype and phenotype data are available through the ALSPAC data team pending a successful application. Access to the ALSPAC data was granted under project # B3136. Summary statistics generated in the current study, including a total of 6 LGC GWASs and 18 cross-sectional GWASs, will be made available on GWAS catalog (https://www.ebi.ac.uk/gwas).

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AUTHOR CONTRIBUTIONS

JM conceived, designed the study, and acquired funding; WQD and KB designed the statistical models, analyzed data, and produced visualizations; WQD designed and carried out the genomic analyses; KB carried out phenotypic analyses; JM and MRM acquired data; WQD, KB, and JM drafted the manuscript. All authors discussed results, critically reviewed the content, provided revisions of the manuscript for important intellectual content, and approved the final version.

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COMPETING INTERESTS

JM is a principal in Beam Diagnostics, Inc. and a consultant to Clairvoyant Therapeutics, Inc. No other authors have disclosures.

ETHICS STATEMENT

Ethical approval for the study was obtained from the ALSPAC Ethics and Law Committee and the Local Research Ethics Committees. Informed consent for the use of data collected via questionnaires and clinics was obtained from participants following the recommendations of the ALSPAC Ethics and Law Committee at the time. All PATH CANN participants underwent informed consent, and the registry was approved by the Hamilton Integrated Review Ethics Board (#1074).

ADDITIONAL INFORMATION

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